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Pulse Programmer for EPR: The New Generation

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[*] [&]

Overview

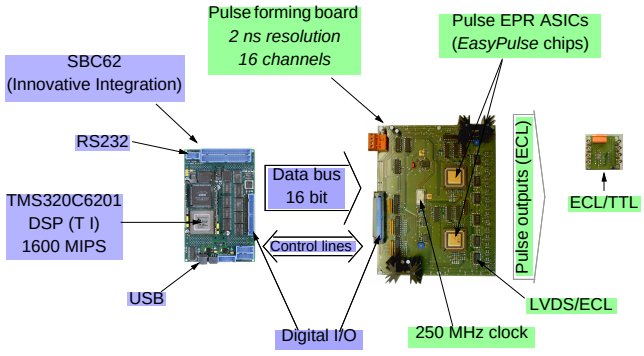
Design criteria

- Versatile sequence control
- High time resolution
- Short rearm time
- High channel density

Applied solution

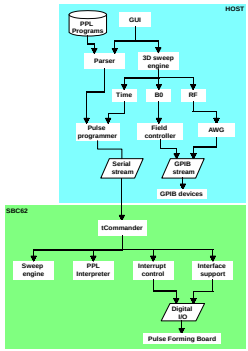
- Pulse EPR Application Specific Integrated Circuit
- Fast Digital Signal Processor
- Pulse Programming Language

Hardware



The pulse forming board provides electrical and logical compatibility between SBC62 supercontroller, EasyPulse chips and the consumer devices. The Voltage Differential Signalling (LVDS) pulses generated by the chips are converted to ECL standard on the board and, if required, to TTL using additional converters.

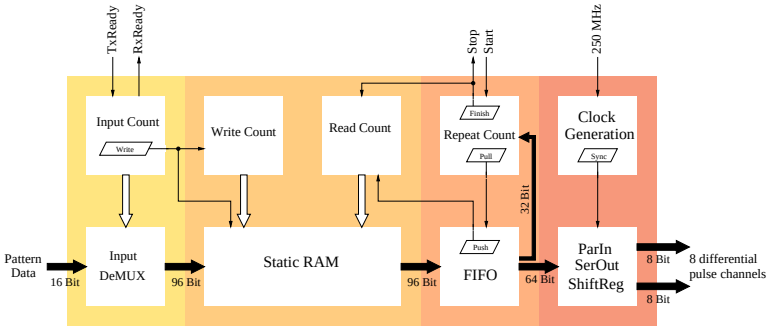
Control program



The host part is a modified version of the program used by the Weizmann and ETH EPR groups [1,2]. The key part of the DSP code is the *interpreter* of the programs written in ***P**ulse **P**rogramming **L**anguage* (*PPL*).

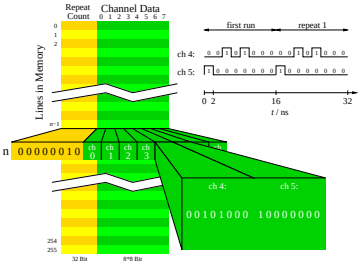
EasyPulse Chip

Block Diagram



Process technology: Austria Micro System 0.6 µm, triple metal, double poly, CMOS n-well, "CUP"
Physical Data: 3202 standard cells; 2 RAM blocks; 14.6 mm^2 die area
Timing Data: 250 MHz main clock; 250 MHz and 62.5 MHz internal clocks; 500·10^6 events/s

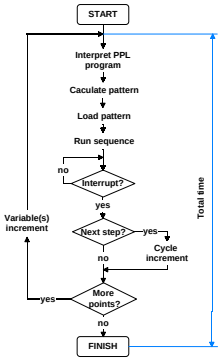
Pattern Coding



The RAM retains 256 lines of 96-bit words. Each word represents 64 bits for the pattern and 32 bits for the number of the pattern repetitions. In turn, the word is composed from the individual pattern of the 8 output channels. 1536 write cycles are required to reload the RAM.

Rearm Test

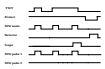
Testing program

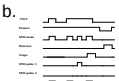


The test is devoted to measure the minimal time at which the pulse experiment can be repeated technically.

Reloading Times

Sequence	Used memory [lines]	Dwell time [ns]	Sweep points	Total time[d] [µs]	Total time / points*cycles [µs]	T_{rep}[e] [µs]
3-pulse ESEEM[a]	15-16	16 2	10^3	1873139 1865248	468 466	460-480
HYSCORE[b]	21-22	2	10^3 x 10^3	2532959115 (42.2 min)	633	630-640
Burst saturation & recovery[c]	117		1			6220

a. 
4-step cycle

b. 
4-step cycle

c. See ➔

d. Measured using the SBC62 timer

e. Measured with an oscilloscope

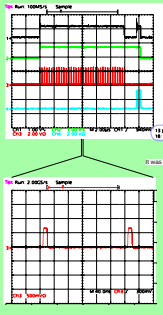
The test demonstrates that standard experiments such as 3-pulse ESEEM and HYSCORE can be repeated at a rate of 2 kHz. The rearm time does not depend on the sweep direction, the next point or the next cycle. Roughly half of the rearm time is used for the data loading to the chips.

Example

```
const time t1 = 12ns ; pulse in burst
time  t = 1.5 us     ; burst length
const time t90 =  100 ns ; pi/2 pulse
const time tau =  300 ns ; delay
const time t180= 200 ns ; pi pulse
int i = 0            ; pulse counter
const int n = 50 ; number of pulses
const int duty = 20 ; duty cycle

program satRecBurst

  trigger 1
  for i from 0 to n do begin
      mwpulse t1
      wait (duty-1)*t1
  end
  wait t1 + t
  mwpulse t90, channel = [1, 2]
  wait tau
  trigger 2
  mwpulse t180, channel = [1, 2]
  wait tau
  trigger 3
  detect re=[1, -1]
```



The pulse program written in *PPL* and the corresponding pulse sequence produced by the pattern generator.

Conclusion

The pulse EPR specific integrated circuits, together with a modern and fast DSP and pulse programming language allow one to program pulse sequences in a easy way and perform EPR experiments in a time which is limited only by the sample properties.

1. J. Shane et al., RSI, 69(9) 3357 (1998)
2. I. Gromov et al., JMR, 149, 196 (2001)